



BASIC CIVIL ENGINEERING

**DR. ANEESH MATHEW
ASSISTANT PROFESSOR
DEPARTMENT OF CIVIL ENGINEERING
NIT TIRUCHIRAPPALLI**



UNIT – I

CIVIL ENGINEERING MATERIALS

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- ▶ **Civil engineering** is a professional engineering discipline that deals with the design, construction, and maintenance of the physical and naturally built environment, including public works such as roads, bridges, canals, dams, airports, sewerage systems, pipelines, structural components of buildings, and railways.

Branches of Civil Engineering

- ▶ STRUCTURAL ENGINEERING
- ▶ CONSTRUCTION ENGINEERING
- ▶ ENVIRONMENTAL ENGINEERING
- ▶ GEOTECHNICAL ENGINEERING
- ▶ TRANSPORTATION ENGINEERING
- ▶ WATER RESOURCES ENGINEERING
- ▶ SURVEYING

STRUCTURAL ENGINEERING

- ▶ Structural engineering is concerned with the structural design and structural analysis of buildings, bridges, towers, flyovers, tunnels, off shore structures like oil and gas fields in the sea, and other structures.
- ▶ This involves identifying the loads which act upon a structure and the forces and stresses which arise within that structure due to those loads, and then designing the structure to successfully support and resist those loads.
- ▶ The loads can be self weight of the structures, other dead load, live loads, moving (wheel) load, wind load, earthquake load, load from temperature change etc.

CONSTRUCTION ENGINEERING

- ▶ This involves planning and execution of the designs from transportation, site development, hydraulic, environmental, structural and geotechnical engineers.
- ▶ As construction firms tend to have higher business risk than other types of civil engineering firms, many construction engineers tend to take on a role that is more business-like in nature: drafting and reviewing contracts, evaluating logistical operations, and closely monitoring prices of necessary supplies.

ENVIRONMENTAL ENGINEERING

- ▶ This deals with the treatment of chemical, biological and/or thermal waste, the purification of water and air, and the remediation of contaminated sites, due to prior waste disposal or accidental contamination.
- ▶ Among the topics covered by environmental engineering are pollutant transport, water purification, waste water treatment, air pollution, solid waste treatment and hazardous waste management.

GEOTECHNICAL ENGINEERING

- ▶ This concerned with the rock and soil that civil engineering systems are supported by.
- ▶ Knowledge from the fields of geology, material science and testing, mechanics, and hydraulics are applied by geotechnical engineers to safely and economically design foundations, retaining walls, and similar structures.

TRANSPORTATION ENGINEERING

- ▶ This concerned with moving people and goods efficiently, safely, and in a manner conducive to a vibrant community.
- ▶ This involves specifying, designing, constructing and maintaining transportation infrastructure which includes streets, canals, highways, rail systems, airports, ports, and mass transit.
- ▶ It includes areas such as transportation design, transportation planning, traffic engineering, some aspects of urban engineering, pavement engineering, Intelligent Transportation System (ITS), and infrastructure management.

WATER RESOURCES ENGINEERING

- ▶ Water resources engineering is concerned with the collection and management of water.
- ▶ As a discipline it therefore combines hydrology, environmental science, meteorology, geology, conservation, and resource management. This area of civil engineering relates to the prediction and management of both the quality and the quantity of water in both underground (aquifers) and above ground (lakes, rivers, and streams) resources.
- ▶ Water resource engineers analyze and model very small to very large areas of the earth to predict the amount and content of water as it flows into, through, or out of a facility

SURVEYING

- ▶ Surveying is the process by which a surveyor measures certain dimensions that generally occur on the surface of the Earth. Surveying equipment, such as levels and theodolites, are used for accurate measurement of angular deviation, horizontal, vertical and slope distances.
- ▶ With computerization, electronic distance measurement (EDM), total stations, GPS surveying and laser scanning have supplemented (and to a large extent supplanted) the traditional optical instruments.
- ▶ This information is crucial to convert the data into a graphical representation of the Earth's surface, in the form of a map. This information is then used by civil engineers, contractors and even realtors to design from, build on, and trade, respectively.



Thank You